

Tutorial 2

- With reference to **Figure 1-1**, assume that **Open Shortest Path First version 2 (OSPFv2)** with Process ID **88** and Area **0** is being implemented to establish dynamic routing between **ROUTER_A**, **ROUTER_B**, and **ROUTER_C**. A **static default route** is pre-configured on **ROUTER_A** to forward traffic to the **ISP**. Apply **OSPFv2** theoretical concepts to answer the following:

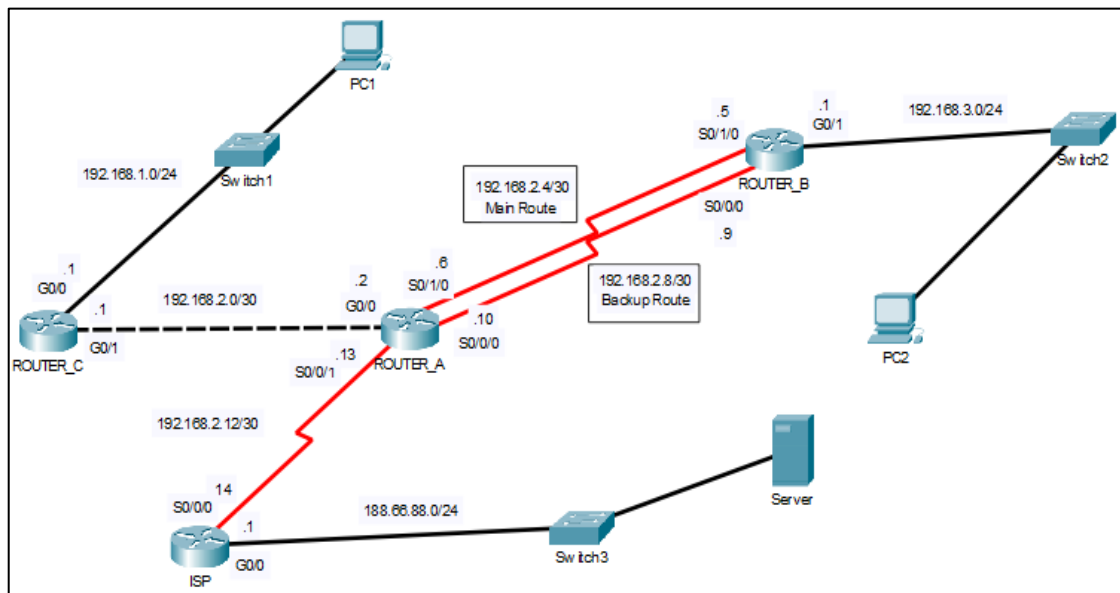


Figure 1-1: A network topology

- As shown in **Figure 1-1**, the connection between **ROUTER_A** and **ROUTER_C** uses the subnet mask **255.255.255.252**. Apply the mathematical rule for deriving the **OSPFv2** wildcard mask from the given subnet mask. Demonstrate your calculation and explain how **OSPFv2** applies this resulting wildcard mask to activate the correct interface for **Area 0**. (5 marks)
- ROUTER_A** uses the **default-information originate** command to propagate its static default route into the **OSPFv2 Area 0** domain. Explain the **TWO (2)** technical requirements that must be met for **ROUTER_A** to successfully advertise this route. (4 marks)

- Differentiate a static routing and OSPF dynamic routing in terms of the following features

Features	Static Routing	OSPF Dynamic Routing
Scalability or usage		
Administrative Effort		
Adaptation to Network Changes		
Router Processing		
Security		
Convergence Time		
Maintenance		
Administrative distance		

3. Refer to the OSPF configuration shown in Figure 3-1 and the network topology shown in Figure 3-2, answer the following questions:

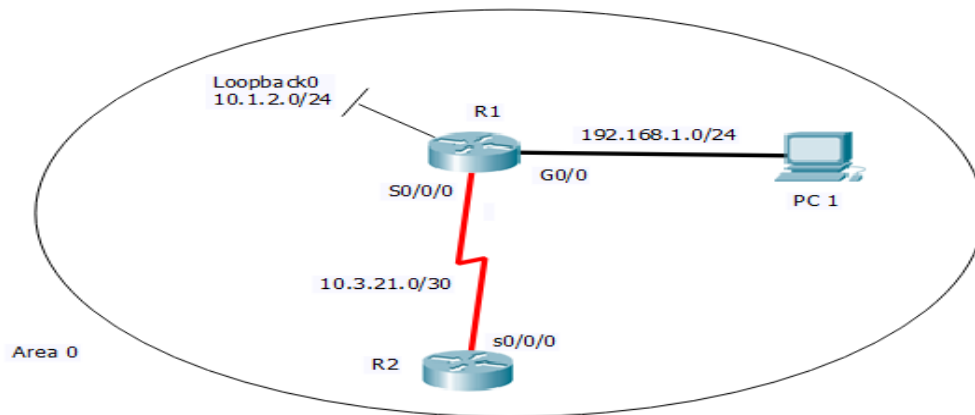


Figure 3-2: A network topology

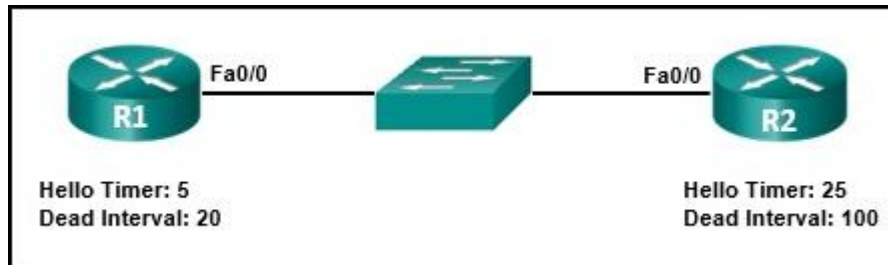
```

R1# show running-config
<output omitted>
interface loopback0
ip address 10.1.2.1 255.255.255.0
!
interface GigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
!
interface Serial0/0/0
ip address 10.3.21.1 255.255.255.252
clock rate 2000000
!
router ospf 1
 network 10.1.2.0 0.0.0.255 area 0
 network 192.168.1.0 0.0.0.255 area 0
 network 10.3.2.0 0.0.0.255 area 0
!
  
```

Figure 3-1: OSPF configuration

- a. Describes the order of precedence on how a router derives the router ID.
 - b. Determine the router ID for R1 in Figure 3-1. Justify your answer.
 - c. Refer to both Figure 3-1 and Figure 3-2, identify and rectify any OSPF configuration error(s).
 - d.
 - (i) Refer to the network topology shown in Figure 3-1, which router interfaces are recommended to set passive interface.
 - (ii) Provide **TWO (2)** reasons why it is recommended to implement passive interfaces.
4. Multiaccess networks can create two challenges for OSPF regarding the flooding of LSAs. What are these challenges? How to solve these challenges?

5. a) Differentiate Hello and Dead interval.
- b) Based on the following diagrams, illustrate the results of configuring hello and dead intervals.



6. With reference to the Figure 4-1, answer the following questions.

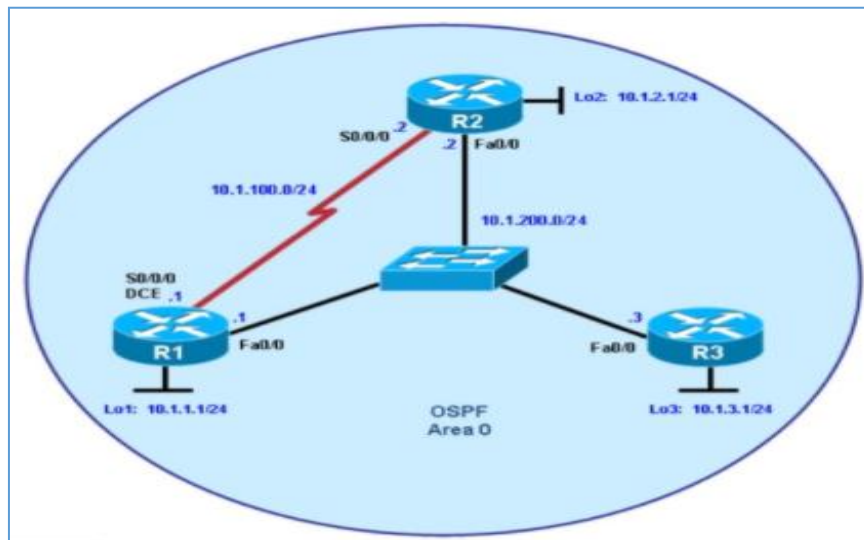


Figure 4-1: An OSPF network

- (i) What is the default interface priority for R1, R2 and R3?
- (ii) What is the router ID for R1, R2 and R3?
- (iii) Which router are DR, BDR and DROthers?
- (iv) Modify the interface priorities to make R1 a DR and R3 a BDR.